

The Centralizer Is Trivial: Closing the Last Construction-Specific Link in the Type III₁ Verdict

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Abstract. The companion paper's type III₁ verdict for the squeeze-thermal state rested on one link cited rather than derived: passing from the computed modular spectrum $\text{Spec}(\Delta_\omega) = [0, \infty)$ to Connes' invariant $S(M)$, which is an intersection over *all* faithful normal states and could in principle be smaller. We close that link for the specific construction. Main theorem: the centralizer of the squeeze-thermal state is trivial, $M_\omega = \mathbb{C}1$. The proof runs through the GNS Liouvillean: a centralizer element's action on the thermal vector produces an invariant vector of the implementing unitary group; in the Araki-Woods form of a quasi-free KMS state that group is generated by $d\Gamma(h \oplus (-h))$; and since h has purely absolutely continuous spectrum with no zero eigenvalue — forced by the Mellin conjugacy, proved in the companion paper — no n -particle sector contains an invariant vector, leaving only the thermal vector itself. Three consequences follow at once: the center satisfies $Z(M) \subseteq M_\omega$, so factoriality is *derived* rather than assumed; Connes' Théorème 3.2.1 applies in its strongest form, since the only nonzero centralizer projection is 1, giving $S(M) \cap \mathbb{R}_+^\times = \text{Spec}(\Delta_\omega) \cap \mathbb{R}_+^\times = \mathbb{R}_+^\times$ and hence type III₁ outright; and the III₁/III_λ fork acquires a second face — in the lattice counterfactual, zero is an eigenvalue of multi-particle Liouvillians (atom of exact mass $\frac{1}{3}$ computed for the grading ladder), so the alternative branch has a nontrivial centralizer, while the forced continuum has none. Supporting verifications: the one-particle covariance bound $\frac{1}{2} \coth(\beta p/2) \geq \frac{1}{2}$ holds exactly (upgrading the state's existence at the one-particle level to elementary status), thermal correlations decay under current smearing ($0.54 \rightarrow 1.6 \times 10^{-4}$), and the infrared condition is *necessary* — without it the two-point function diverges logarithmically, with increments matching $\beta^{-1} \ln 100$ to four decimals. After this note, every external input to the verdict is textbook-level.

1. The gap, stated precisely

Connes' invariant is $S(M) = \bigcap_\varphi \text{Spec}(\Delta_\varphi)$, the intersection over all faithful normal (semifinite) weights. The companion paper computed $\text{Spec}(\Delta_\omega) = [0, \infty)$ for the squeeze-thermal state $\omega = \omega_\beta$: an *upper* bound on $S(M)$ only. The conversion of one-state data into the invariant is Connes' Théorème 3.2.1 [Connes 1973]: $S(M) \cap \mathbb{R}_+^\times = \Gamma(\sigma^\varphi)$, where $\Gamma(\sigma^\varphi) = \bigcap \{ \text{Spec}(\Delta_{\varphi_e}) : e \in \text{Proj}(M_\varphi), e \neq 0 \}$ — an intersection over reductions by nonzero projections *in the centralizer* $M_\varphi = \{x : \sigma_t^\varphi(x) = x \ \forall t\}$. If the centralizer is trivial, the only such projection is 1, the intersection collapses to the single term, and $S(M) \cap \mathbb{R}_+^\times = \text{Spec}(\Delta_\varphi) \cap \mathbb{R}_+^\times$ with no further hypotheses. This note proves the centralizer of ω_β is trivial.

2. The centralizer is the fixed-point algebra of the dynamics

Lemma 1 (standard). For a β -KMS state ω with respect to a dynamics τ_t , the modular flow is the dynamics reparametrized, $\sigma_t^\omega = \tau_{-\beta t}$; hence M_ω is exactly the fixed-point algebra of τ . [Takesaki; Bratteli–Robinson II.]

For the squeeze-thermal state, τ is the Bogoliubov flow of e^{ith} with $h = -i\partial_u$ on the chiral scale line — which is, by the companion paper's Proposition 1, the squeeze flow itself in Mellin coordinates.

3. Main theorem

Theorem. $M_{\omega_\beta} = \mathbb{C}1$.

Proof. Let $x \in M_{\omega_\beta}$, and let $(\mathcal{H}, \Omega, U(t))$ be the GNS data, with $U(t)$ the unitary group implementing τ and fixing Ω . From $\tau_t(x) = x$ and $U(t)\Omega = \Omega$: $U(t)x\Omega = \tau_t(x)\Omega = x\Omega$ — the vector $x\Omega$ is U -invariant.

The GNS representation of a quasi-free KMS state has the Araki-Woods form [standard: Araki-Woods; Bratteli–Robinson II]: $\mathcal{H}$ is the Fock space over the doubled one-particle space $H_1 \oplus \bar{H}_1$, Ω its vacuum, and $U(t) = \Gamma(e^{it\ell})$ with one-particle Liouvillean $\ell = h \oplus (-h)$ — particles and holes.

By the companion paper's Proposition 1, $\text{Spec}(h) = [0, \infty)$, purely absolutely continuous, with no zero eigenvalue; hence ℓ has purely absolutely continuous spectrum and no eigenvalues. On the n -particle sector ($n \geq 1$), the spectral measure of $d\Gamma(\ell)$ is an n -fold convolution in which every factor is absolutely continuous — and the convolution of an absolutely continuous measure with any measure is absolutely continuous [standard measure theory]. So no sector with $n \geq 1$ contains an eigenvector; in particular none contains a U -invariant vector. The only invariant vectors lie in the vacuum sector: $x\Omega \in \mathbb{C}\Omega$, say $x\Omega = c\Omega$. Since ω_β is faithful, Ω is separating, and $x = c1$ with $c = \omega_\beta(x)$. ■

The absolute continuity is doing all the work, and it was *forced*, not chosen: the Mellin conjugacy fixed the spectrum, and the spectrum fixed the centralizer.

4. Three consequences

Corollary 1 (factoriality, derived). Modular automorphisms fix the center pointwise, so $Z(M) \subseteq M_{\omega_\beta} = \mathbb{C}1$: M is a factor. What the companion paper had to assume as part of the standard package is now a one-line consequence of the theorem.

Corollary 2 (the link, closed). With $M_{\omega_\beta} = \mathbb{C}1$, Théorème 3.2.1 gives $S(M) \cap \mathbb{R}_+^\times = \text{Spec}(\Delta_{\omega_\beta}) \cap \mathbb{R}_+^\times = \mathbb{R}_+^\times$, using the companion paper's computed spectrum. By Connes' trichotomy, M is of type III_1 . The verdict no longer rests on any construction-specific citation.

Corollary 3 (canonicity). Granting hyperfiniteness of the quasi-free algebra [standard for this class; flagged], Haagerup's uniqueness theorem identifies M as *the* hyperfinite type III_1 factor.

Remark. That a faithful normal state with trivial centralizer can live only on a type III_1 factor is also a known theorem of the recent ergodic-states literature (surfaced earlier in this series' searches alongside the Connes–Størmer homogeneity results); it corroborates Corollary 2 from an independent angle but is not load-bearing here — the direct chain above suffices.

5. The second face of the fork

The $\text{III}_1/\text{III}_\lambda$ dichotomy of the companion papers reappears in the centralizer itself. In the lattice counterfactual — energies on the grading ladder $(1, 2, 3)E_0$ — the two-particle Liouvillean value $p_1 - p_2$ has an *atom at zero* of exact mass $3 \times (\frac{1}{3})^2 = \frac{1}{3}$: genuine invariant multi-particle vectors exist, and the centralizer is nontrivial (for the Powers-type states it is famously large). In the forced continuum case, the mass of $\{|p_1 - p_2| < \epsilon\}$ vanishes linearly — computed: $\text{mass}/\epsilon \rightarrow 2.000$ as $\epsilon \downarrow 0$ — no atom, no invariant vectors, trivial centralizer. The fork is thus visible three independent ways: in the generated spectral group (companion paper), in the state-space geometry (homogeneity versus invariants), and now in the centralizer (trivial versus large).

6. Supporting verifications

Existence, upgraded. With the chiral restriction the one-particle generator satisfies $h > 0$, and the thermal covariance is $S(p) = \frac{1}{2} \coth(\beta p/2)$. Verified: $S(p) - \frac{1}{2} \geq 0$ exactly on the grid, with equality approached in the vacuum limit $p \rightarrow \infty$ — the quasi-free positivity (uncertainty) bound holds pointwise, so the state exists by the elementary quasi-free construction; the companion paper's "cited pending" existence status is upgraded at the one-particle level.

Mixing, illustrated. With current smearing $\hat{f}(p) = p e^{-p^2/2}$, the thermal correlation function decays cleanly: $|c(t)| = 0.540, 0.056, 2.5 \times 10^{-3}, 1.6 \times 10^{-4}$ at $t = 0, 5, 20, 80$ — the dynamical face of the theorem (mixing implies trivial fixed points; the Liouvillean proof is the sharper route, but the decay makes it visible).

The infrared condition is necessary, quantitatively. Without the p -factor in the smearing ($\hat{f}(0) \neq 0$), the equal-time two-point integral diverges logarithmically: truncating at $\epsilon = 10^{-2}, 10^{-4}, 10^{-6}$ gives 4.358, 8.963, 13.568, with increments 4.6051 and 4.6052 against the predicted $\beta^{-1} \ln 100 = 4.6052$. The current-type smearing of the construction is not a convenience; it is exactly what makes the state's correlations finite.

7. Scope

After this note, the external inputs to the type III_1 verdict are all textbook-level, listed exhaustively: the KMS-modular identity (Lemma 1); the Araki-Woods GNS form of quasi-free KMS states; Connes' Théorème 3.2.1 (statement verified against the literature in this series' sessions); hyperfiniteness of the quasi-free algebra (standard for this class, still cited rather than re-derived); and Haagerup's uniqueness. Construction-specific work is complete: spectrum (companion paper, proved), generated group (computed),

centralizer (this note, proved), factoriality (derived). Still open, unchanged: uniqueness of ω_β among $SO(3)$ -invariant squeeze-KMS states, and the full infrared execution of the field construction — §6 settles that the infrared condition is necessary and what it must accomplish, not the complete functional-analytic write-out. One adjacency for the record, found during verification: for dilation-invariant quantum field theories it is a known result that the spectra of all modular operators coincide with $\mathbb{R}_+$ and that central decompositions contain only type III_1 factors — an independent, physics-side precedent for exactly the dilation-to- III_1 mechanism realized here.

References

- Companion papers: *The Squeeze-Thermal State: A Type-I No-Go, the Scale-Line Construction, and the Type III_1 Verdict*; *The Second Contraction: From Algebraic R-Flux to the Heisenberg Algebra, the Dilation Symmetry It Creates, and the $SO(3)$ Remnant of G_2* .
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